

Name: _____

AS Chemistry Unit 1

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2016					
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2016	16	22	29	36	43	57
2017	13	20	28	36	44	60
2018	18	24	31	38	45	59
2019	17	23	30	37	44	58
2022	13	20	28	36	44	60
2023	12	18	24	31	38	52

Surname	Centre Number	Candidate Number
Other Names		2



GCE AS/A Level

2410U10-1 – **NEW AS**



S16-2410U10-1

CHEMISTRY – Unit 1

The Language of Chemistry, Structure of Matter and Simple Reactions

A.M. FRIDAY, 27 May 2016

1 hour 30 minutes

Section A

Section B

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.

Section B Answer **all** questions in the spaces provided.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11(c)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 7.	10	
8.	16	
9.	11	
10.	17	
11.	13	
12.	13	
Total	80	

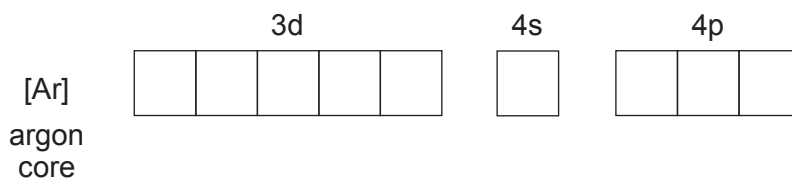
2410U101
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MAY162410U10101

SECTION A*Answer all questions in the spaces provided.*

1. Using the convention of arrows to represent electrons, complete the electronic structure of an arsenic atom, As. [1]



2. Give the oxidation number of molybdenum in the ion MoO_4^{2-} . [1]

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3. The half-life of the radioactive isotope oxygen-15 is 2 minutes.

- (a) Give the atomic number, mass number and symbol of the nuclide produced when 1 atom of ^{15}O decays by the emission of one positron. [1]

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- (b) If a quantity of ^{15}O decays by positron emission, determine what fraction of ^{15}O is left after 10 minutes. [1]

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4. Explain why the second molar ionisation energy of sodium is greater than its first molar ionisation energy. [1]

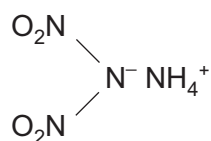
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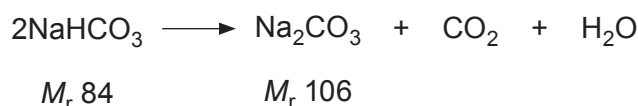


5. The explosive ADN has the structural formula shown below.



Give the empirical formula of ADN. [1]

6. Sodium carbonate, Na_2CO_3 , is made by heating sodium hydrogencarbonate, NaHCO_3 .

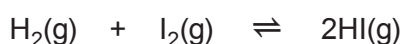


Calculate the atom economy of this reaction. [1]

Atom economy = %

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7. Hydrogen and iodine react together to give hydrogen iodide.



- (a) Write the expression for the equilibrium constant in terms of concentration, K_c , for this reaction. [1]

- (b) The equilibrium constant, K_c , for this reaction has a value of 46.0 at a certain temperature. Calculate the equilibrium concentration of hydrogen at this temperature. The equilibrium concentration of iodine is 1.20 mol dm^{-3} and the equilibrium concentration of hydrogen iodide is $15.00 \text{ mol dm}^{-3}$. [2]

Concentration of hydrogen = mol dm^{-3}

10



SECTION B

Answer all questions in the spaces provided.

8. (a) Iron(III) oxide, Fe_2O_3 , and carbon monoxide react to give iron and carbon dioxide.

Give the equation for this reaction. Explain why it is described as a redox process. [2]

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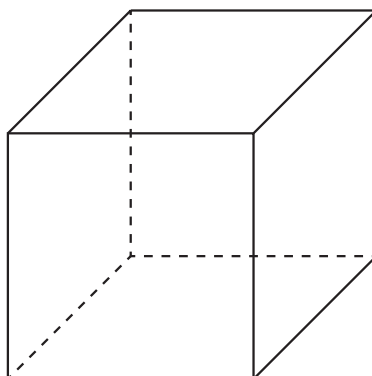
- (b) The iron obtained from the blast furnace contains some sulfur as an impurity. This is removed during Basic Oxygen Steelmaking (BOS), by adding magnesium, which removes the sulfur as magnesium sulfide, MgS .

350 tonnes of impure iron used for BOS contain 0.02 % of sulfur. Calculate the mass, in **kilograms**, of magnesium needed to remove all the sulfur. [2]

Mass = kg

- (c) Magnesium sulfide has the same crystal structure as sodium chloride.

Use the diagram below to show the crystal structure of magnesium sulfide, clearly labelling the formula of each species present. [2]



- (d) Magnesium sulfide reacts with water producing gaseous hydrogen sulfide and magnesium hydroxide.



- (i) State what would be seen during the reaction apart from gas bubbles. [1]

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- (ii) A student added a small sample of magnesium sulfide of mass 0.224 g to some water in a fume cupboard.

Calculate the maximum volume, in cm^3 , of hydrogen sulfide produced at 25°C . [3]

Volume = cm^3

- (e) When barium sulfide is added to water a similar reaction occurs to that described in part (d) with gaseous hydrogen sulfide and barium hydroxide being the products.

State with a reason, how the observation for this reaction would be different from that seen with magnesium sulfide. [2]

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- (f) Barium hydroxide can also be produced by adding barium oxide to water.

Give the equation for this reaction and estimate the pH of the product. [2]

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(g) Explain why it is unlikely that stable compounds containing Ba^{3+} ions can exist.

[2]

Examiner
only

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9. (a) Ethanedioic acid (oxalic acid) has the formula $(\text{COOH})_2$.

It can be made in the laboratory by oxidising sugar with a powerful oxidising agent (shown as [O] in the equation below).

- (i) Balance the equation. [1]



- (ii) After purification, the oxalic acid is produced as white crystals of a hydrate, $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. On heating this hydrate, water is lost, leaving anhydrous oxalic acid.



In an experiment 5.67 g of the hydrate were heated to constant mass, giving 4.05 g of the anhydrous acid.

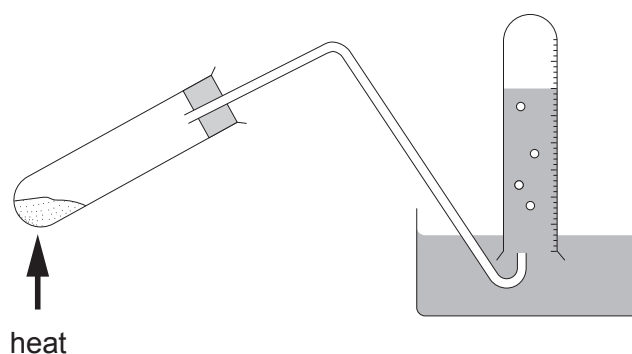
Calculate the value of x in $(\text{COOH})_2 \cdot x\text{H}_2\text{O}$. You **must** show your working. [3]

$x = \dots\dots\dots$

- (b) Another method of producing oxalic acid in the laboratory is to heat potassium methanoate, HCOOK . This produces potassium oxalate, which is then acidified to give the acid.



Some students carried out this experiment using the apparatus below. They collected the hydrogen gas produced over water.



A few students obtained a smaller volume of hydrogen than expected.

Suggest **two** practical reasons why this may have occurred, apart from leaks. [2]

1

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(c) Gareth was asked to analyse a mixture of potassium methanoate and potassium oxalate. He weighed out 4.69 g of the mixture and carried out the method below.

- The mixture was completely dissolved in distilled water.
- Calcium chloride solution was added to this solution (only calcium oxalate was precipitated).
- The mixture was filtered and the precipitated calcium oxalate washed and dried.

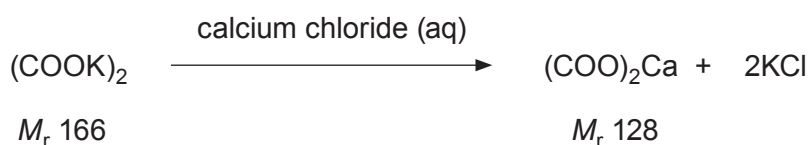
(i) Suggest how Gareth would know when enough calcium chloride solution had been added to react with all of the potassium oxalate present. [2]

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(ii) In his experiment Gareth obtained 2.49 g of pure dry calcium oxalate (M_r 128).



Calculate the percentage of potassium oxalate in the mixture, giving your answer to the appropriate number of significant figures. [3]

Percentage of potassium oxalate = %



10. (a) The mass spectrum of a silicon sample showed the presence of the three isotopes ^{28}Si , ^{29}Si and ^{30}Si . The percentage of ^{28}Si present was 92.2% and the mass spectrum showed that the percentage of ^{29}Si present was twice that of ^{30}Si .

Calculate the relative atomic mass of this sample of silicon.

[3]

You **must** show your working.

Relative atomic mass =

- (b) Rhiannon studied the properties of silicon and found that its structure is similar to that of diamond. In her report she stated that

- the Si–Si–Si bond angle is 109.5°
- silicon is a poor conductor of electricity
- each silicon atom is bonded to four other silicon atoms.

- (i) State the name of the shape that has this bond angle.

[1]

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- (ii) Explain why solid silicon is a very poor electrical conductor.

[1]

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- (iii) Explain why the bonding between each atom is covalent.

[1]

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- (c) Hydrofluoric acid, HF, is supplied as a solution that contains 50 % by mass of hydrogen fluoride. The density of this solution is 1.17 g cm^{-3} .

Calculate the concentration of this solution in mol dm^{-3} . [2]

Concentration = mol dm^{-3}

- (d) (i) Hydrofluoric acid will dissolve silica, SiO_2 , to produce hexafluorosilicic acid. This acid contains the SiF_6^{2-} ion.

Use the information from the table below to draw the shape of the SiF_6^{2-} ion, showing the F–Si–F bond angle. Give a reason for your answer. [3]

Ion	Number of bonding electron pairs	Number of lone electron pairs on the central silicon atom
SiF_6^{2-}	6	0

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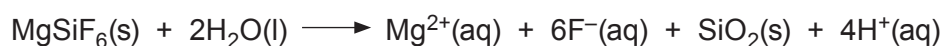
- (ii) Hexafluorosilicic acid, H_2SiF_6 , (M_r 144) can be added to drinking water to promote good dental health. When added to water all the fluorine in the acid is available as fluoride ions.

The water in an area of Derbyshire contains 0.15 mg dm^{-3} of fluoride ions.

Calculate how much hexafluorosilicic acid, in mg, should be added to each dm^3 of water to increase the fluoride level to 0.76 mg dm^{-3} . [3]

Mass of hexafluorosilicic acid = mg

- (e) Solid magnesium hexafluorosilicate hydrolyses rapidly when added to water. One equation for this reaction is as follows.



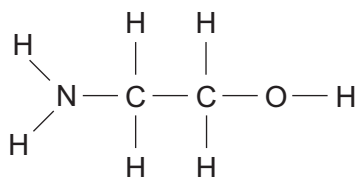
In an experiment 2.60 g of magnesium hexafluorosilicate were added to 1.00 dm^3 of water.

Calculate the pH of the resulting mixture. [3]

pH =



11. (a) 2-Aminoethanol is a colourless liquid.



2-aminoethanol

It is very soluble in water owing to its ability to hydrogen bond with water molecules.

Complete the diagram above to show how 2-aminoethanol is able to hydrogen bond to water molecules. [4]

- (b) The boiling temperatures of 1,2-diaminoethane and 2-aminoethanol are shown in the table.

Compound	Formula	Boiling temperature / °C
1,2-diaminoethane	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$	117
2-aminoethanol	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{OH}$	170

Use these figures to comment on the strength of the intermolecular forces between the molecules in each compound suggesting reasons for your answer. [3]

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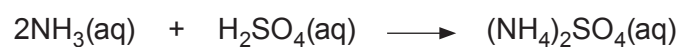
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- (c) Ammonium sulfate can be prepared in the laboratory by neutralising aqueous sulfuric acid with ammonia solution in an acid-base titration.



Outline a method to obtain pure dry crystals of ammonium sulfate by this procedure.

[6 QER]

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12. (a) Some students were given a mixture made up of about 50 % each of sodium chloride and sodium iodide and asked to design tests to show the presence and concentration of iodide ions in this mixture.

- (i) One student decided to try electrolysis. He dissolved some of the mixture in distilled water to give a concentrated solution. He then passed electricity through the solution, using inert electrodes. The negative ions moved to the anode where they lost electrons.

State what was seen at the anode to confirm the presence of iodide ions.
Explain your answer, including a half-equation.

[3]

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- (ii) Another student decided to use a chemical test. She dissolved some of the mixture in distilled water and then added a little aqueous silver nitrate. After noting what was seen, she added some aqueous ammonia to the mixture and shook it.

State and explain what occurred during this test.

[3]

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- (iii) In a further test the concentration of the iodide ions present was found by a titration with potassium iodate(V) dissolved in a strong acid. 11.24 g of the mixture of sodium chloride and sodium iodide were dissolved in distilled water and made up to 250 cm^3 . The iodide ions present in 25.0 cm^3 of this solution required 18.00 cm^3 of the potassium iodate(V) solution of concentration 0.100 mol dm^{-3} , to completely react.

Calculate the number of moles of iodide ions (present as sodium iodide, M_r 150) in 25.0 cm^3 of the solution and hence the exact percentage of sodium iodide in the mixture. [3]

(1 mol of potassium iodate(V) reacts with 2 mol of sodium iodide)

Percentage of sodium iodide = %

- (b) Hydrogen chloride is a colourless gas that absorbs at 278 nm in the ultraviolet region of the electromagnetic spectrum.

Use this information and the data sheet to show that the energy of the H–Cl bond is 431 kJ mol^{-1} . [2]



(c) The table shows bond energy values and absorption maxima for the hydrogen halides.

H–X	Bond energy / kJ mol^{-1}	Absorption maximum / nm
H–F	562	212
H–Cl	431	278
H–Br	366	326
H–I	299	400
H–At		

The accepted range for wavelengths for the visible region of the electromagnetic spectrum is 400–700 nm.

Suggest and explain why it is probable that H–At would be a coloured gas. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2


GCE AS/A LEVEL – NEW

2410U10-1



S17-2410U10-1

CHEMISTRY – AS unit 1
The Language of Chemistry, Structure of Matter
and Simple Reactions

FRIDAY, 26 MAY 2017 – MORNING

1 hour 30 minutes

Section A

Section B

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 5.	10	
6.	20	
7.	15	
8.	17	
9.	18	
Total	80	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.

Section B Answer **all** questions in the spaces provided.

 Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

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The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

 The assessment of the quality of extended response (QER) will take place in **Q.9(a)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



MAY172410U10101

SECTION A

*Answer **all** questions in the spaces provided.*

1. Using **outer** electrons only, draw a dot and cross diagram to show the bonding in calcium bromide. [2]

2. Identify the **two** elements from the following list that together produce the most ionic bond. Explain your choice.

bromine

magnesium

oxygen

sodium

[2]

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3. (a) State how a coordinate bond differs from a covalent bond. [1]

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- (b) Give **one** example of a species containing a coordinate bond. [1]

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4. There are 33 known isotopes of krypton. One of the isotopes – ^{81}Kr – decays by electron capture.

(a) Write an equation to show this decay. [1]

- (b) Krypton-81 has been used for dating old groundwater. It takes 6.87×10^5 years for 2.0 g of ^{81}Kr to decay to 0.25 g. Calculate its half-life. [1]

Half-life = years

5. Boron has a relative atomic mass of 10.8. It has only two naturally-occurring isotopes, one of which has an abundance of 80.0% and a mass of 11. Calculate the mass of the second isotope. [2]

Mass =



SECTION B

Answer all questions in the spaces provided.

6. (a) The atomic spectrum of hydrogen consists of several separate series of lines.
- (i) In the Balmer series, when an electron returns from the third shell to the second shell, a red line is seen at a wavelength of 656 nm. Calculate the energy of the radiation emitted at this wavelength. [3]

Energy = J

- (ii) Explain how the Lyman series can be used to calculate the ionisation energy of hydrogen. [2]

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- (b) State and explain how you would expect the ionisation energy of hydrogen to compare with the first ionisation energy of:

- (i) helium [2]

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- (ii) lithium [2]

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- (c) (i) On adding a strip of magnesium to hydrochloric acid, 125 cm^3 of hydrogen gas formed at a temperature of 24°C and a pressure of $1.01 \times 10^5\text{ Pa}$. Calculate the volume occupied by this gas under the same pressure and at a temperature of 52°C . [2]

Volume = cm^3

- (ii) In another experiment, 160 cm^3 of hydrogen formed at 20°C and $1.01 \times 10^5\text{ Pa}$. Assuming that hydrogen behaves as an ideal gas, calculate the amount, in moles, of hydrogen formed. [3]

n = mol



- (d) Hydrogen can form a range of covalent hydrides such as water, H_2O , and beryllium hydride, BeH_2 . A student said that since the ratio of hydrogen to the other element is the same in both compounds, the shapes of the two compounds will be the same. Is she correct? Justify your answer. [3]

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- (e) (i) On forming ice at 0°C , 50.0 cm^3 of water expands to occupy 54.5 cm^3 . Calculate the density of ice at 0°C . [2]

Density = g cm^{-3}

- (ii) Explain why ice and water have different densities at 0°C . [1]

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7. (a) A student is asked to prepare a standard solution of sodium carbonate of concentration $0.0500 \text{ mol dm}^{-3}$.

The first step involved in the preparation of a standard solution is to weigh the appropriate mass of solid in a weighing bottle.

Calculate the mass of sodium carbonate required and describe the remaining steps that the student should take to prepare 250 cm^3 of this standard solution. [6]

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- (b) The standard solution prepared in part (a) was used to determine the concentration of dilute hydrochloric acid.

25.0 cm³ samples of the sodium carbonate solution were titrated against the hydrochloric acid. These were the results.

Titration	1	2	3	4
Final reading / cm ³	26.50	26.80	26.20	26.55
Initial reading / cm ³	0.40	0.15	0.00	0.25
Titre / cm ³	26.10	26.65	26.20	26.30

- (i) Calculate the mean titre that should be used to determine the concentration of the hydrochloric acid. [1]

Mean titre = cm³

- (ii) The burette used in the titrations has an uncertainty for each reading of $\pm 0.05\text{cm}^3$. Estimate the maximum percentage error in the **titre** in titration 4. Show your working. [1]

Percentage error = %

- (iii) Apart from errors in reading the burette, suggest **one** reason why incorrect titres may have been obtained when carrying out the titrations. Explain the effect of this error on the value of the titre obtained. [2]

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- (c) A student wanted to verify that the ammonium sulfate content in a fertiliser was as given on the packet.

He boiled 4.24 g of the fertiliser with 50.0 cm³ of 1.00 mol dm⁻³ sodium hydroxide solution until no more ammonia was evolved.

The residual solution containing excess sodium hydroxide was made up to 250 cm³ with distilled water in a standard volumetric flask. A 25.0 cm³ portion of this solution required 22.65 cm³ of 0.090 mol dm⁻³ hydrochloric acid for neutralisation.

- (i) Calculate the number of moles of sodium hydroxide in the 250 cm³ flask and hence the number of moles that reacted with the fertiliser. [3]

$n(\text{NaOH}) = \dots\dots\dots \text{mol}$

- (ii) The equation for the reaction between ammonium sulfate and sodium hydroxide is:

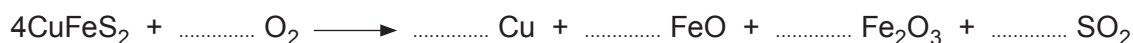


Use your answer to part (i) to calculate the percentage by mass of ammonium sulfate in the fertiliser. [2]

Percentage = $\dots\dots\dots\%$



8. (a) Copper can be extracted from the mineral chalcopyrite. The mineral is smelted by heating with air. The equation for this reaction is as follows.



- (i) Balance the equation above. [1]
- (ii) Give the full electronic configuration of a copper atom. [1]

- (iii) A sample of rock contains 1.30 % by mass of chalcopyrite. Assuming this is the only source of copper in the rock, calculate the percentage by mass of copper in the sample. [2]

Percentage = %

- (iv) During the smelting Cu_2S is produced. This is collected and blown with air to produce copper.



This is classified as a redox reaction. Use oxidation numbers to explain which elements have been oxidised and which reduced. [3]

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- (b) Sulfur dioxide can react with oxygen and water to form sulfuric acid. Although sulfuric acid is a strong acid, it does not have to be a concentrated acid.

Explain the difference between the terms *strong acid* and *concentrated acid*. [2]

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- (c) (i) A student was given an aqueous solution of an acid HX and was asked to find out if it was a strong or weak acid. 25.0 cm^3 of the acid HX required 15.90 cm^3 of $0.0125\text{ mol dm}^{-3}$ sodium hydroxide solution for complete neutralisation.

Calculate the concentration of the acid. Assume that HX reacts with the sodium hydroxide in a 1:1 molar ratio. [2]

Concentration = mol dm^{-3}

- (ii) A teacher measured the pH of the aqueous solution and found it to be 2.10. She told the student that the acid in the solution must be a strong one. Is she correct? Justify your answer. [2]

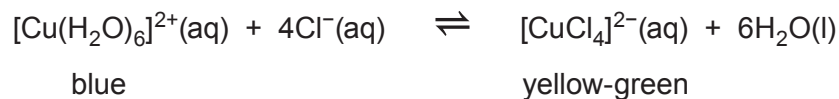
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- (d) The commonest complex ion that copper forms in solution is the hexaaquacopper(II) ion, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$. If concentrated hydrochloric acid is added to a solution containing hexaaquacopper(II) ions a new complex forms as the following equilibrium is established.



The forward reaction is endothermic.

- (i) State and explain what you would observe on adding water to the equilibrium mixture. [2]

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- (ii) State and explain what you would observe on heating the equilibrium mixture. [2]

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9. (a) A student is given five beakers labelled **A**, **B**, **C**, **D** and **E**. Each contains a different solution. The solutions are barium bromide, barium nitrate, calcium chloride, magnesium chloride and magnesium nitrate but it is not known which beaker contains which solution.

She is also given solutions of three reagents: silver nitrate, sodium hydroxide and sulfuric acid.

Devise a plan that uses the reagents to unambiguously determine which solution is in which beaker.

You should include the observations that enable you to positively identify **all** five solutions.
[6 QER]

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- (b) Give a test which would confirm the presence of calcium ions in aqueous calcium chloride. State the result of the test. [2]

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- (c) A solution is known to contain a mixture of sodium carbonate and sodium nitrate only. To determine the carbonate ion concentration in the mixture, aqueous barium chloride was added in excess. The precipitate produced was filtered off and dried by strong heating.

- (i) Write the **ionic** equation, including state symbols, for this precipitation reaction. [1]

- (ii) State why the aqueous barium chloride was added in excess. [1]

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- (d) All Group 2 carbonates are insoluble and can be precipitated, dried and weighed in experiments similar to that in part (c). However, the strong heating needed to drive off all the water can cause a problem in accurately determining the mass of the carbonate precipitated. Suggest what this problem might be. [2]

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(e) Explain the following observations by reference to the bonding present in each of the substances.

(i) Ionic substances such as calcium chloride can conduct electricity under certain circumstances. [2]

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(ii) Iodine is a solid which vaporises on gentle warming. [2]

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(iii) A metal such as magnesium is malleable. [2]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		2

**GCE AS/A LEVEL**

2410U10-1



S18-2410U10-1

CHEMISTRY – AS unit 1**The Language of Chemistry, Structure of Matter and Simple Reactions**

TUESDAY, 22 MAY 2018 – MORNING

1 hour 30 minutes

Section A

Section B

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.11(a)(i)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 6.	10	
7.	12	
8.	15	
9.	15	
10.	13	
11.	15	
Total	80	

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SECTION A

Answer all questions in the spaces provided.

1. For the ionic compound caesium chloride, state the coordination number of the chloride ion. [1]

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2. The first four ionisation energies for an element are shown in the table below.

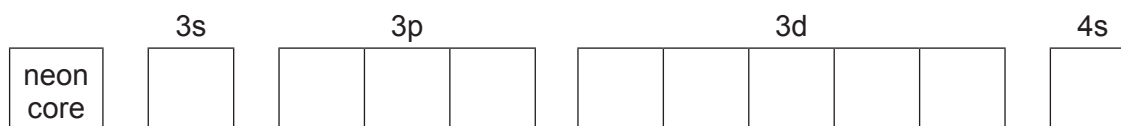
Ionisation energy / kJ mol ⁻¹			
1st	2nd	3rd	4th
548	1060	4120	5440

State to which group in the Periodic Table the element belongs.

[1]

.....

3. By inserting arrows to represent electrons, complete the electronic structure of a chromium atom. [1]



4. Phosphorus(V) chloride reacts with water to form phosphoric acid and hydrogen chloride. The reactants and products are shown in the equation below but the equation is **not** balanced.



- (a) Balance the equation. [1]
- (b) Calculate the atom economy for the formation of phosphoric acid in this reaction. [2]

Atom economy = %

5. (a) Give the meaning of the term *electronegativity*. [1]

.....

- (b) Explain why electronegativity increases across a period in the Periodic Table. [1]

.....

6. Bromine is produced commercially from the bromide ions in sea water by reaction with chlorine.

- (a) Give the **ionic** equation for this reaction. [1]

.....

- (b) Describe a test, apart from the use of chlorine, to show that a solution contains bromide ions. Give the reagent(s) and observation(s). [1]

Reagent(s)

Observation(s)



SECTION B

Answer all questions in the spaces provided.

7. (a) Melting temperatures vary down groups and across periods.

(i) Explain why chlorine is a gas but iodine is a solid at room temperature and pressure. [3]

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(ii) Explain why sodium has a lower melting temperature than aluminium. [1]

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(iii) Explain why silicon has a higher melting temperature than phosphorus. [1]

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(b) State and explain how you would expect the first ionisation energy of nitrogen to compare with the first ionisation energy of oxygen. [2]

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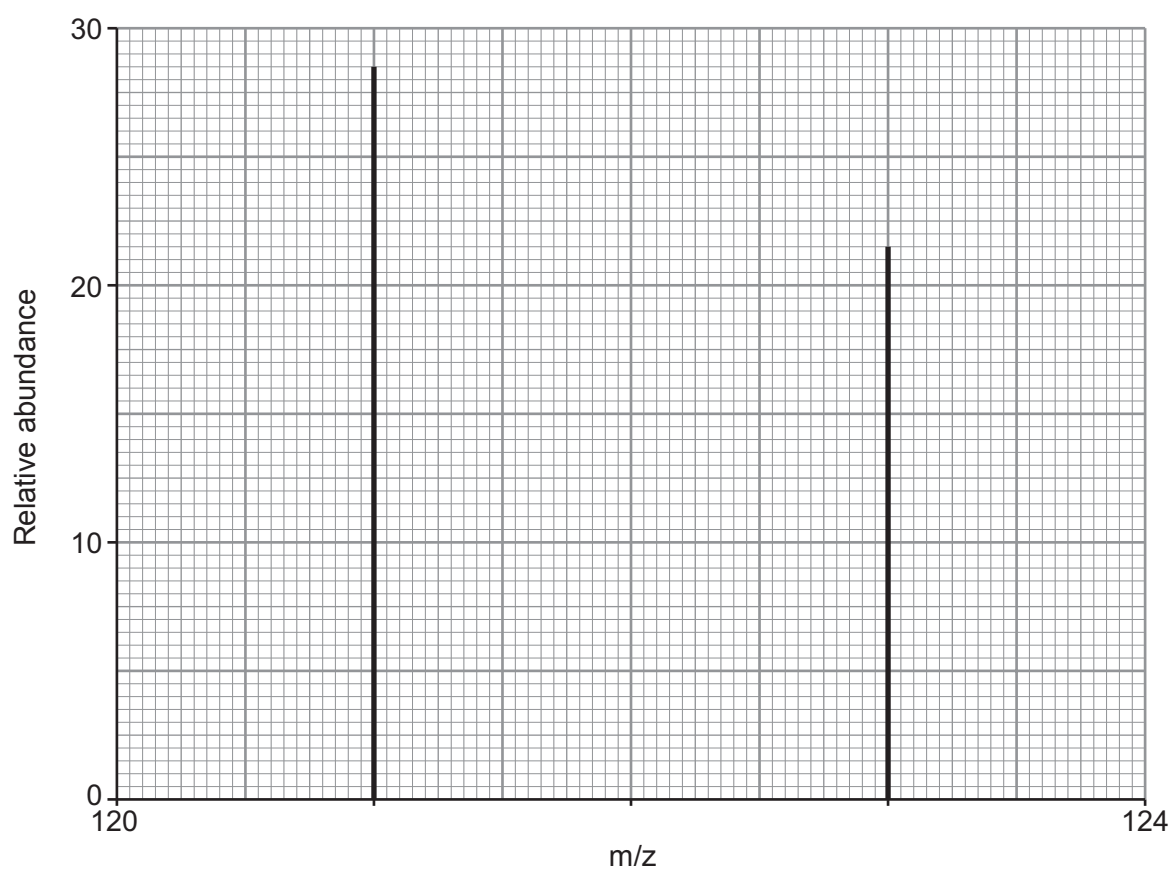
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(c) Antimony is in Group 5. Its mass spectrum shows that it has two stable isotopes.



Calculate the relative atomic mass of antimony. You **must** show your working.

[2]

Relative atomic mass =



(d) Although radiation from radioisotopes is harmful to health many beneficial uses of radioactivity have been found.

The table below gives some information about four radioactive isotopes.

Isotope	Radiation emitted	Half-life
^{90}Sr	β	28 years
^{99}Tc	γ	6 hours
^{210}At	α	8.1 hours
^{228}Th	α	1.9 years

Use **all** the data given to choose which isotope is the most suitable to use as a tracer in medicine. Explain your answer. [3]

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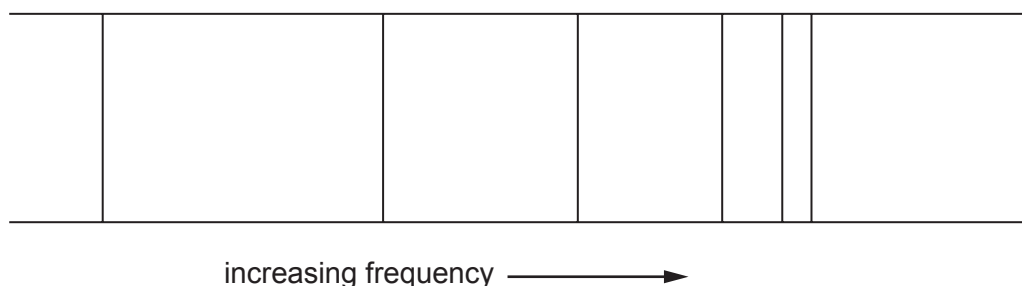
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8. (a) The diagram below shows part of the atomic emission spectrum of hydrogen.



- (i) Use the letter **A** to label the line of longest wavelength on the diagram. [1]
- (ii) Explain why hydrogen atoms emit only certain definite frequencies of visible light. [2]

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.....

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- (b) The ionisation energy of a hydrogen atom is 2.18×10^{-21} kJ.

- (i) Explain what this statement means. [2]

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.....

- (ii) Calculate the minimum frequency of radiation required to ionise a hydrogen atom in its ground state. [3]

Frequency = s^{-1}



- (c) Hydrazine is a compound of hydrogen and nitrogen only. It is a colourless, flammable liquid which was used in various rocket fuels.

0.160 g of hydrazine on vaporisation at 398 K and 1 atm pressure has a volume of 163 cm³.

Calculate its volume at 273 K and 1 atm pressure and hence show that its molecular formula is N₂H₄. [3]

.....
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- (d) (i) Draw a dot and cross diagram to show the electron arrangement in hydrazine, N₂H₄. Show outer electrons only. [2]

- (ii) Hydrazine contains polar covalent bonds between nitrogen and hydrogen atoms. State what is meant by a *polar* covalent bond. [1]

.....
.....

- (e) Hydrazine acts as a base in a similar way to ammonia.

Suggest an equation for the equilibrium formed when hydrazine dissolves in water. [1]

.....



9. Ascorbic acid, $C_6H_8O_6$, is the main component of vitamin C tablets. Its name is derived from *a-* (meaning “no”) and *scorbutus* (scurvy), the disease caused by a deficiency of vitamin C. A student was asked to find the percentage of ascorbic acid in identical vitamin C tablets.

She was told to use the following method.

- Fill a burette with $0.100 \text{ mol dm}^{-3}$ sodium hydroxide solution.
- Weigh a conical flask and record its mass.
- Add a vitamin C tablet to the flask, reweigh it and record its mass.
- Add about 50 cm^3 of deionised water to the flask and swirl to break up the tablet.
- Heat the flask gently for 5 to 10 minutes.
- After the solution has cooled add a few drops of a suitable indicator.
- Carry out a rough titration of this solution with the sodium hydroxide solution.
- Accurately repeat the procedure several times and calculate a mean titre.

- (a) A **three** decimal place balance was used. The mass of each vitamin C tablet was 500 mg.

Calculate the maximum percentage error in the weighing of the tablet.
You **must** show your working.

[2]

Maximum percentage error = %

- (b) (i) Suggest why she did not need to measure the volume of water accurately. [1]

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- (ii) Suggest why she heated the flask for 5 to 10 minutes. [1]

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.....



- (c) The student used the results from three titrations to calculate a mean titre. Some of her results are shown below.

Titration	1	2	3
Final reading / cm ³	26.90	26.90	
Initial reading / cm ³	0.25	0.15	0.20
Titre / cm ³	26.65	26.75	

Mean titre = 26.73 cm³

Determine the **final reading** for the third titration.

[2]

Final reading = cm³

- (d) Ascorbic acid can decompose upon exposure to air. If this reaction occurred before the titration was completed, state how it might affect the titration results. Explain your answer. [2]

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- (e) The equation for the reaction between ascorbic acid and sodium hydroxide is given below.



M_r 176

The percentage of ascorbic acid is identical in each 500 mg tablet. Calculate the percentage of ascorbic acid in each vitamin C tablet. [3]

Percentage ascorbic acid = %

- (f) Sulfuric acid and hydrochloric acid are strong acids.

- (i) Calculate the pH of a solution of $0.010 \text{ mol dm}^{-3}$ sulfuric acid, H_2SO_4 . [2]

pH =

- (ii) When hydrochloric acid is heated with MnO_2 it reacts according to the following equation.



Explain why this can be classified as a redox reaction. [2]

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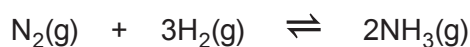
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10. (a) Nitrogen can react with hydrogen to form ammonia.



A mixture of N_2 and H_2 is left to react at a certain temperature, until it reaches equilibrium. The equilibrium mixture has the following composition.

N_2	1.16 mol dm^{-3}
H_2	1.60 mol dm^{-3}
NH_3	$0.752 \text{ mol dm}^{-3}$

- (i) A student said that the equilibrium must lie to the left because the concentrations of nitrogen and hydrogen are greater than that of ammonia. Is he correct?

Justify your answer by calculating a value for K_c for this equilibrium.
Give the unit for K_c .

[4]

 $K_c = \dots\dots\dots$ Unit $\dots\dots\dots$

$\dots\dots\dots$
 $\dots\dots\dots$

- (ii) When the temperature is increased the equilibrium yield of NH_3 decreases. The student said that the reaction is endothermic. Is he correct?

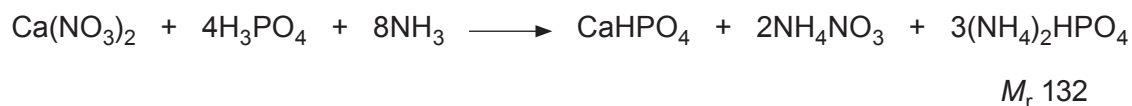
Justify your answer by using Le Chatelier's principle.

[2]

$\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$



- (b) Ammonia can be used as part of the nitrophosphate process to produce the fertiliser diammonium hydrogenphosphate (DAP) which has the formula $(\text{NH}_4)_2\text{HPO}_4$.



Calculate the maximum mass of DAP, in kg, that could be made from 1.00 tonne of ammonia. [3]

Maximum mass = kg

- (c) Calculate the volume, in cm^3 , that 2.54×10^{-3} mol of nitrogen occupies at a temperature of 120°C and a pressure of 101 kPa. [4]

Volume = cm^3



Examiner
only

11. (a) A solution contains one cation and up to three different anions. The anions possibly present are carbonate, chloride and sulfate.

(i) Devise a plan that unambiguously proves which anions are present in the mixture.

You should also give any observations and conclusions that enable you to identify the anions. [6 QER]

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- (ii) A student said that the cation in the mixture can only be sodium since all the possible anions form a soluble salt with sodium. Is he correct?

Justify your answer and state how you could prove if the statement were true. [2]

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- (b) A student is given four materials and asked to identify the **type of structure** present in each one by carrying out a series of tests.

She is told that the maximum temperature of a Bunsen burner flame is about 800 °C. She is also told that in at least one case, it will **not** be possible to come to a definite conclusion.

Her results are shown in the table below.

	A	B	C	D
Melting temperature / °C	100	>800	>800	>800
Solubility in water	soluble	insoluble	insoluble	soluble
Conductivity of solid	none	none	good	none
Conductivity of solution	none			good

- (i) Use the information in the table to identify each type of structure.
Where a definite conclusion cannot be reached explain your reasoning. [4]

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QUESTION CONTINUES ON PAGE 18



Examiner
only

(ii) For **one** of the materials where the type of structure could not be identified, suggest what further test(s) are needed to identify the type of structure. [2]

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(iii) Suggest why it is difficult to identify a material as a metal when it is in powdered form. [1]

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15

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2

**GCE AS/A LEVEL**

2410U10-1



S19-2410U10-1

CHEMISTRY – AS unit 1**The Language of Chemistry, Structure of Matter and Simple Reactions**

MONDAY, 20 MAY 2019 – MORNING

1 hour 30 minutes

Section A**Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.**Section B** Answer **all** questions in the spaces provided.Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.12(c)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 8.	10	
9.	16	
10.	14	
11.	12	
12.	14	
13.	14	
Total	80	

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SECTION A*Answer all questions in the spaces provided.*

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in magnesium fluoride. [2]

2. In some areas, fluoride ions are added to drinking water.

State **one** benefit of adding fluoride ions to water. [1]

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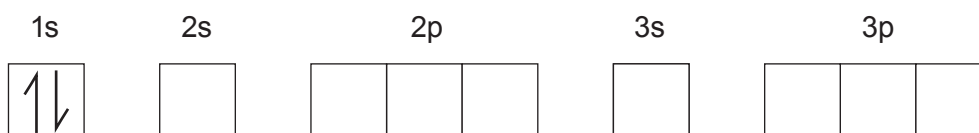
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3. Complete the following definition of relative atomic mass. [1]

The relative atomic mass of an element is the average mass of one atom of the element relative to

.....

4. Sodium forms only one stable ion. By inserting arrows to represent electrons, complete the electronic structure of this ion. [1]



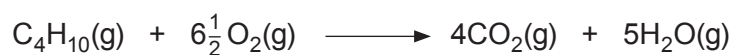
5. Give the oxidation number of vanadium in VOCl_3 . [1]

.....

6. Phosphoric acid has the formula H_3PO_4 . Write the formula of magnesium phosphate. [1]

.....

7. Cooking fuel for outdoor camping contains butane, which reacts with oxygen according to the following equation.



If 1.00 mol of butane reacts in this way, calculate the number of **molecules** of carbon dioxide that will be formed. [1]

Molecules of CO_2 =

8. 9.60 g of titanium reacts completely with 3.68 dm^3 of oxygen gas at 298 K and 1 atm to form an oxide.

Calculate the empirical formula of this oxide. [2]

Empirical formula



SECTION B

Answer all questions in the spaces provided.

9. (a) Consider the elements labelled **A-G**. These are not chemical symbols.

A	$1s^2 2s^2 2p^1$
B	$1s^2 2s^2 2p^3$
C	$1s^2 2s^2 2p^6$
D	$1s^2 2s^2 2p^6 3s^2$
E	$1s^2 2s^2 2p^6 3s^2 3p^1$
F	$1s^2 2s^2 2p^6 3s^2 3p^6$
G	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

- (i) Give the letter (**A-G**) of the element with the largest **first** ionisation energy.

Give **two** reasons for your answer.

[3]

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- (ii) Give the letter (**A-G**) of the element with the largest **last** ionisation energy.

Give a reason for your answer.

[2]

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- (iii) Give the letters (**A-G**) of **all** the elements above that are metals.

[2]

.....

- (iv) Give the letters (**A-G**) of **all** the elements above that form basic oxides.

[1]

.....



(b) Magnesium exists as three naturally-occurring stable isotopes. They can be identified using a mass spectrometer.

(i) State how magnesium ions are formed in a mass spectrometer. [1]

.....

.....

(ii) State how magnesium ions are separated in a mass spectrometer. [1]

.....

.....

(c) Magnesium-28 is an unstable radioactive isotope that decays by β -emission.

(i) Give the mass number and symbol of the element formed as a product of the radioactive decay of magnesium-28. [1]

.....

(ii) If it takes 84 hours for the activity of the isotope to decay to $\frac{1}{16}$ th of its original activity, calculate its half-life. [1]

Half-life = hours



- (d) Magnesium is a typical metal. Describe the bonding in magnesium and explain why it is ductile. You may include a diagram as part of your answer. [3]

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- (e) According to the label on the bottle, the concentration of magnesium ions in a sample of Welsh mineral water is 15 mg / litre.

Calculate the concentration of magnesium ions in mol dm^{-3} . [1]

Concentration = mol dm^{-3}



10. When an electrical discharge passes through gaseous hydrogen at low pressure, electromagnetic radiation is emitted.

- (a) Describe the processes within a hydrogen atom that cause electromagnetic radiation to be emitted. [2]

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- (b) If the electromagnetic radiation in part (a) is passed through a spectrometer, several series of converging lines are observed.

- (i) Explain why there are several series of lines. [1]

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- (ii) Explain why the lines within each series converge. [1]

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- (c) The convergence limit of the Lyman series of lines occurs at a wavelength of 91.2 nm.

- (i) State what the limit represents. [1]

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- (ii) Calculate the energy, in kJ mol^{-1} , of the convergence limit. [4]

Energy = kJ mol^{-1}



- (d) Hydrogen forms when aluminium reacts with sulfuric acid.



- (i) Calculate the volume of hydrogen, in cm^3 , that would be produced if 0.131 g of aluminium were added to an excess of sulfuric acid at a temperature of 25°C and a pressure of 1 atm. [3]

Volume = cm^3

- (ii) Calculate the volume of hydrogen that would be produced if the same experiment were carried out at 50°C and 1.6 atm. [2]

(If you do not have an answer in part (i), assume that the volume is 200 cm^3 . This is **not** the correct answer.)

Volume = cm^3



11. A student was asked to find the percentage of calcium carbonate in a sample of chalk. He used the following chemicals.

- Three chalk pieces of identical composition and mass 2.54 g
- Hydrochloric acid solution of concentration 1.00 mol dm^{-3}
- Sodium hydroxide solution of concentration $0.100 \text{ mol dm}^{-3}$

Method

- Use a burette to measure 50.00 cm^3 of hydrochloric acid (an excess) into a 100 cm^3 beaker.
- Put a piece of chalk into the beaker and leave until the reaction finishes.
- Filter the solution into a conical flask to remove any solid impurities.
- Add a few drops of indicator to the solution in the conical flask and titrate against the sodium hydroxide solution.
- Repeat the procedure using the other chalk pieces and calculate a mean titre.
- Use the mean titre to calculate the percentage of calcium carbonate in the chalk sample.

Results

Mass of each chalk piece = 2.54 g

Titration	1	2	3
Final reading / cm^3	16.80	33.05	16.70
Initial reading / cm^3	0.20	16.80	0.35
Titre / cm^3	16.60	16.25	16.35

$$\text{Mean titre} = \frac{16.60 + 16.25 + 16.35}{3} = 16.40 \text{ cm}^3$$

- (a) State how the student would know that the reaction between the chalk and acid had finished. [1]

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(b) Suggest and explain **two** improvements to the student's method. [4]

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(c) The equation for the reaction between calcium carbonate and hydrochloric acid is as follows.



Use this and the student's results, including the mean titre of 16.40 cm^3 , to calculate the percentage of calcium carbonate in the chalk sample. [4]

Percentage = %



(d) Comment on the validity of the mean titre calculated.

[1]

Examiner
only

(e) Another student follows the same procedure but filters the solution before the reaction is complete. State what effect, if any, this would have on the value of the titre. Justify your answer.

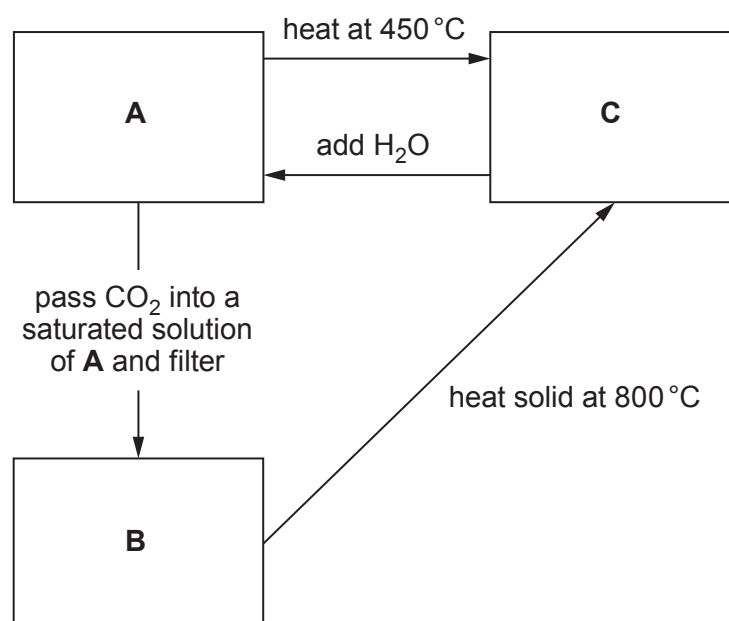
[2]

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12. (a) A series of experiments is performed on three white solids, **A**, **B** and **C**. All contain the same Group 2 metal ion. Compound **A** is the metal hydroxide.

The experiment is summarised below.



- The compounds give a definite colour in a flame test
- Compound **A** is only slightly soluble in water
- Compound **B** is insoluble in water

- (i) A student correctly identified the metal ion as calcium. Give **two** reasons why she came to that conclusion. [2]

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- (ii) Name compound **B**. [1]

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- (iii) 0.110 g of compound **C** was completely neutralised by 26.10 cm³ of 0.150 mol dm⁻³ hydrochloric acid.

Given that 1 mol of compound **C** reacts with 2 mol of acid, calculate the relative formula mass of compound **C** and hence confirm that the metal ion is calcium.

You **must** show your working.

[2]

- (b) Group 2 metals are not the only elements that form 2+ ions.

A sample of an element has two isotopes, one with 70 neutrons and the other 72 neutrons. This element forms a 2+ ion containing 48 electrons. The relative abundances of the isotopes are 57.9% and 42.1% respectively.

Calculate the relative atomic mass of the element. Give your answer to **four** significant figures.

[3]

$A_r =$



- (c) Describe, giving **brief** experimental details, how you would obtain the maximum amount of magnesium carbonate from a known mass of magnesium, in a **two-stage** process. Include appropriate chemical / ionic equations in your answer. [6 QER]

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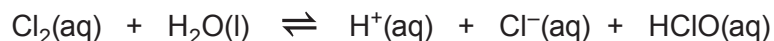
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13. (a) During the 2016 Olympic games, the diving pool had to be closed after the water turned green. A report in the media incorrectly suggested that there was too much chlorine in the pool.

When chlorine gas dissolves in cold water, a pale green solution is formed. In this solution, the following equilibrium is established.



pale green

colourless

- (i) Chemical equilibria are often described as dynamic equilibria.

Explain the term *dynamic equilibrium*.

[1]

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- (ii) Use Le Chatelier's principle to explain why the pale green colour disappears if sodium hydroxide solution is added. [2]

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(b) Chlorine can react directly with metals to form chlorides.

- (i) I. Calcium chloride can exist as an anhydrous salt or as a hydrated salt, $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$.

In an experiment to determine the extent of hydration a sample of hydrated calcium chloride, $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$, with a mass of 3.29 g was heated to remove all water of crystallisation. The solid remaining had a mass of 1.67 g.

Calculate the value of x in the formula $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$.

You **must** show your working.

[3]

$x =$

- II. Suggest how a student doing this experiment would ensure that all the water had been removed. [1]

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- (ii) The melting temperature of sodium chloride is 1074 K but sodium iodide has a melting temperature of 934 K.

Suggest why the melting temperature of sodium iodide is lower than that of sodium chloride. [1]

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.....



(c) Chlorine forms molecules and ions with other halogens.

- (i) While chlorine has a boiling temperature of 238 K, the boiling temperature of iodine monochloride, ICl, is 371 K.

Suggest why the boiling temperature of iodine monochloride is higher than that of chlorine. [2]

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- (ii) Name the shape of the $[\text{ClF}_6]^+$ ion. [1]

.....

- (iii) $[\text{ClF}_2]^+$ and $[\text{ClF}_2]^-$ are two other ions containing a chlorine atom.

A student said that their shapes must be different.

Is he correct? Justify your answer using VSEPR theory. [3]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2

**GCE AS/A LEVEL**

2410U10-1

**TUESDAY, 17 MAY 2022 – MORNING****CHEMISTRY – AS unit 1**

**The Language of Chemistry, Structure of Matter
and Simple Reactions**

1 hour 30 minutes

Section A**Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.

Section B Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10(a)**.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1. to 7.	10	
8.	14	
9.	12	
10.	13	
11.	17	
12.	14	
Total	80	

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SECTION AAnswer **all** questions.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in sodium oxide. [2]

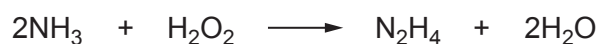
2. Complete the following sentence: [1]

Hydrogen bonding is an intermolecular force that occurs between molecules containing hydrogen atoms bonded to small, atoms which have lone pairs of electrons for example

3. By inserting arrows to represent electrons, show the electronic configuration of a calcium atom. [1]

1s	2s	2p	3s	3p	3d	4s

4. Hydrazine can be manufactured from ammonia.



Calculate the atom economy of this reaction. [1]

atom economy = %



5. The ammonium ion contains a 'coordinate bond'. Explain what is meant by this term. [1]

.....

.....

6. Uranium is used in nuclear fuel reactors. One of its isotopes, uranium-235, has a half-life of 7.03×10^8 years and decays by α -emission.

- (a) Give the mass number and symbol of the element formed as a product of the radioactive decay of uranium-235. [1]

.....

- (b) If a quantity of uranium-235 decays, state what fraction is left after 2.812×10^9 years. [1]

fraction left =

7. Calculate the mass of calcium that contains the same number of atoms as there are molecules in 9.1 g of sulfur dioxide, SO_2 . [2]

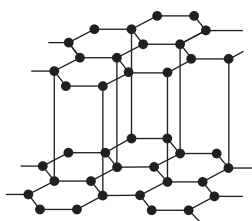
calcium mass = g



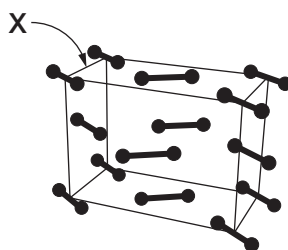
SECTION B

Answer **all** questions.

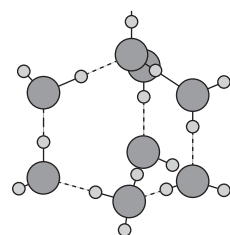
8. (a) The diagrams below represent the structures of cadmium metal, caesium chloride, graphite, ice, iodine and sodium chloride.



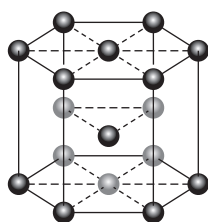
A graphite



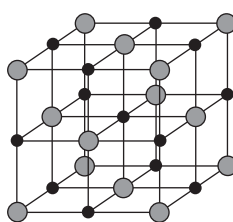
B



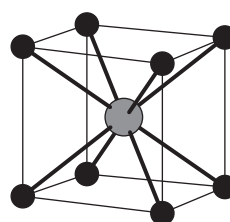
C



D cadmium



E



F

- (i) Label the remaining structures in the spaces provided. [2]
- (ii) Complete the table showing the trend in melting temperatures of the six substances. [2]

Substance	Melting temperature
.....	Lowest
.....	
.....	
caesium chloride	
sodium chloride	
.....	
	Highest



- (iii) For structure **B**, name the type of bond or force represented by the letter **X**. [1]

.....

- (iv) Name the **two** substances which are good electrical conductors in the solid state. [1]

.....

.....

- (v) Explain why graphite is suitable for use in pencils. [2]

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- (vi) Cadmium is a typical metal.
Give a brief description of metallic bonding. [3]
You may include a diagram to support your answer.

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- (b) Many houses have been built on disused industrial sites. A housing developer wants to test whether the soil on a particular site is contaminated with cadmium ions (Cd^{2+}).

They extract a sample and prepare a solution. Cadmium ions behave in the same way as magnesium ions when treated with sodium hydroxide solution and sodium sulfate solution.

- (i) State what the developer would see if cadmium ions were present when they added:

I. sodium hydroxide solution. [1]

.....

II. sodium sulfate solution. [1]

.....

- (ii) Write an ionic equation, including state symbols, for **one** of the above observations. [1]

.....



9. (a) (i) Explain the origins of emission spectra. [2]

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- (ii) Give **one** difference in the appearance of absorption and emission spectra. [1]

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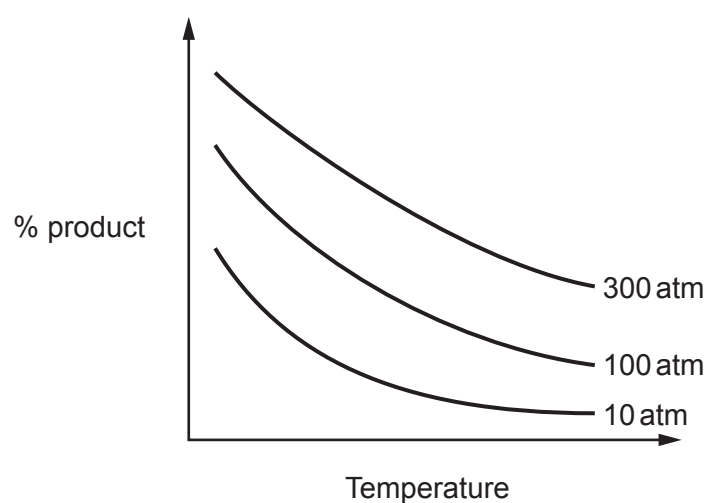
.....

- (b) A line in the emission spectrum of an element has a wavelength of 95.0 nm. Calculate the frequency of this line in megahertz, MHz. [2]

frequency = MHz



- (c) The diagram below shows how the percentage product varies with temperature and pressure for an equilibrium process.



Use the diagram and Le Chatelier's principle to explain whether:

- (i) the forward reaction is endothermic or exothermic.

[2]

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- (ii) the forward reaction involves an increase or decrease in the number of moles of gas.

[2]

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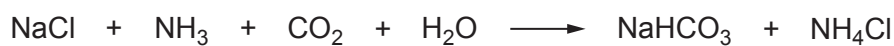
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- (d) The Solvay process is the major industrial process for the manufacture of sodium carbonate. Two of the stages in the process are shown below:



Calculate the maximum mass of sodium carbonate, in kg, which could be obtained from 15.0 tonnes of sodium chloride. Give your answer to an appropriate number of significant figures.

[3]

maximum mass = kg

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10. (a) Group 1 and Group 2 metals are in the s-block of the Periodic Table. Using potassium and calcium as examples, discuss the similarities and differences between Group 1 and Group 2 with respect to:

- the reaction of the metals with cold water.
- the solubility of the carbonates.

[6 QER]

You should include appropriate chemical equations in your answer.

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- (b) The equation for the reaction between potassium carbonate and hydrochloric acid is given below.



A 1.40g sample of impure potassium carbonate was added to excess dilute hydrochloric acid. The impurity is unreactive and only the potassium carbonate reacts with the acid.

The volume of carbon dioxide released was 186 cm^3 when measured at 298 K and $1.01 \times 10^5\text{ Pa}$.

Calculate the mass of the impurity.

[3]

mass of impurity = g

- (c) A solution is thought to contain potassium chloride. Describe suitable tests that a student could do to confirm this. Include the expected observations.

[2]

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- (d) By referring to ionisation energies, explain why stable compounds containing K^{2+} ions are unlikely to form.

[2]

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11. (a) Arsenic oxide, As_2O_3 , is prepared on an industrial scale by roasting arsenic-containing ores such as arsenopyrite, FeAsS , in air. The other products formed are iron(III) oxide and sulfur dioxide.

(i) State the oxidation state of arsenic in As_2O_3 . [1]

.....

(ii) Give a balanced chemical equation for the industrial production of As_2O_3 from FeAsS . [2]

.....

- (b) As_2O_3 is moderately soluble in water. 100 cm^3 of a saturated solution at 25°C contains 2.06 g .

When dissolved in water, the oxide reacts to form arsenous acid.



- (i) Calculate the concentration of the arsenous acid, in mol dm^{-3} , in the saturated solution. [3]

concentration of $\text{H}_3\text{AsO}_3 = \dots\dots\dots \text{mol dm}^{-3}$

- (ii) A solution of arsenous acid has a pH of 5.11.

Calculate the hydrogen ion concentration of this solution. [1]

$[\text{H}^+] = \dots\dots\dots \text{mol dm}^{-3}$



- (c) The formula for arsenous acid can be written as $\text{As}(\text{OH})_3$ since it contains three hydroxyl (OH) groups bonded to arsenic.

Suggest the shape around the arsenic atom in $\text{As}(\text{OH})_3$. Justify your answer by using VSEPR theory. [3]

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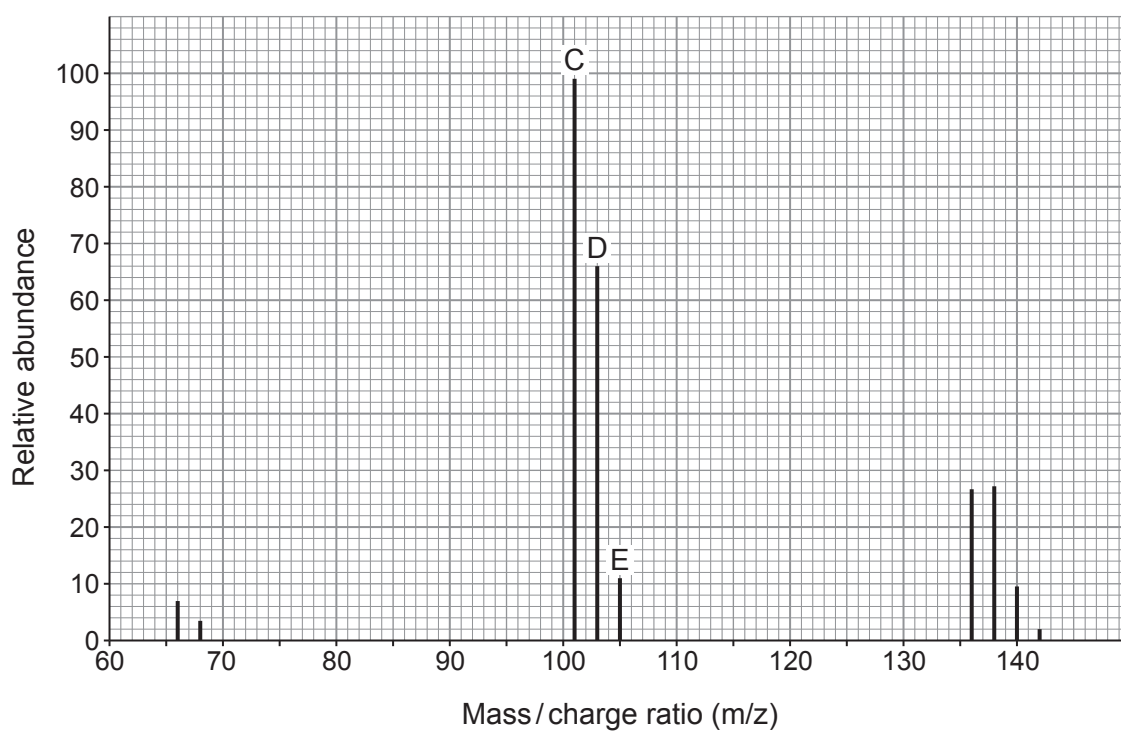
- (d) Phosphorus can form two chlorides, PCl_3 and PCl_5 .

0.181 g of a chloride of phosphorus gave 39 cm^3 of vapour at 1 atm pressure when heated to 87°C . The sample was completely vapourised.

Show that the chloride of phosphorus was PCl_3 . [4]



- (e) The molecular ion region of the mass spectrum of PCl_3 , is shown below.



- (i) Identify the species responsible for peak C at m/z 101.

[1]

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- (ii) Explain why the height ratio of peaks C:E is 9:1.

[2]

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12. A student was asked to find the identity of a Group 1 metal carbonate by titration.

He was told to use the following method.

- Weigh a sample of the carbonate in a weighing bottle.
- Transfer the carbonate into a beaker and weigh the bottle afterwards.
- Add water to the beaker to dissolve the carbonate.
- Transfer the solution to a volumetric flask.
- Add more water to make the final volume 250.0 cm^3 of solution.
- Accurately transfer 25.0 cm^3 of this solution into a conical flask.
- Add 2–3 drops of a suitable indicator to this solution.
- Fill a burette with 0.100 mol dm^{-3} hydrochloric acid solution.
- Carry out a rough titration of the carbonate solution with the hydrochloric acid.
- Accurately repeat the titration until you get concordant titres and calculate a mean titre.

(a) Another student said that there were two errors in making the 250.0 cm^3 carbonate solution.

Error 1: A small amount of solid remained in the weighing bottle.

Error 2: A small amount of solution remained in the beaker.

Comment on the suggested errors.

If the student is correct suggest how the method could be improved.

If the student is incorrect, explain why.

[2]

Error 1

.....

.....

Error 2

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.....

(b) State why he adds an indicator to this solution.

[1]

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.....

(c) Suggest why he was told to carry out a rough titration first.

[1]

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Examiner
only

(d) State what you understand by the term ‘concordant titres’. [1]

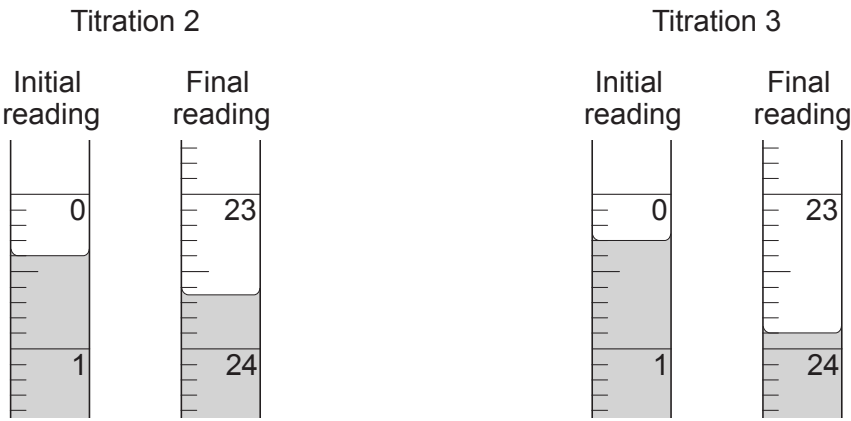
(e) Some of the student’s results are shown below:

Mass of weighing bottle + carbonate /g	13.73
Mass of weighing bottle /g	12.48

Titration	Rough	1	2	3
Final reading /cm ³	24.20	23.70		
Initial reading /cm ³	0.00	0.10		
Titre /cm ³				

concentration of hydrochloric acid = 0.100 mol dm⁻³

The diagrams below show the initial burette reading and the final burette reading for the second and third titrations.



Complete the student’s table and calculate the mean titre. [3]

mean titre = cm³



- (f) The equation for the reaction between the metal carbonate and hydrochloric acid is given below. **M** represents the symbol of the Group 1 metal.



- (i) Calculate the number of moles of M_2CO_3 in 25.0 cm^3 of the solution. [2]

number of moles =

- (ii) Calculate the relative formula mass of the carbonate and hence deduce the Group 1 metal in the carbonate.

You **must** show your working.

[4]

group 1 metal =

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2

**GCE AS/A LEVEL**

2410U10-1



S23-2410U10-1

TUESDAY, 16 MAY 2023 – MORNING**CHEMISTRY – AS unit 1****The Language of Chemistry, Structure of Matter and Simple Reactions**

1 hour 30 minutes

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions.**Section B** Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.7(a)**.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A 1. to 6.	10	
Section B 7.	10	
8.	12	
9.	12	
10.	15	
11.	21	
Total	80	

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SECTION AAnswer **all** questions.

1. Complete the table below.

[2]

Molecule	Number of bonding pairs of electrons in outer shell	Number of lone pairs of electrons in outer shell	Shape
BeCl ₂	2	0	
PCl ₃			pyramidal

2. Complete the table below to show the type or types of bonding present in the following solids.

[2]

Solid	Type or types of bonding
calcium	
iodine	

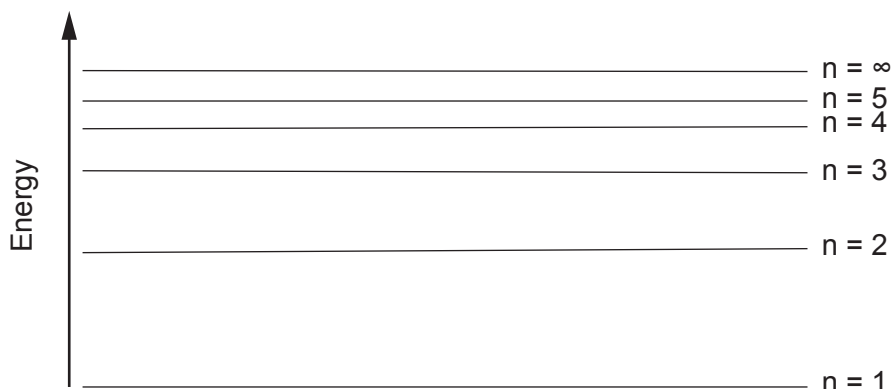
3. Give the oxidation number of rhenium in ReOCl₄.

[1]

.....



4. The diagram below shows the electron energy levels for a hydrogen atom.



- (a) On the diagram, draw an arrow to represent the transition corresponding to the ionisation of the atom. Label this arrow **A**. [1]
- (b) On the diagram, draw an arrow to represent the transition corresponding to the first line in the visible region in the atomic spectrum. Label this arrow **B**. [1]

5. A student said that $^{32}_{16}\text{S}^{2-}$ and $^{40}_{20}\text{Ca}^{2+}$ have the same electronic configuration.

Do you agree? Give a reason for your answer. [1]

.....

.....

6. Copper can be extracted from copper(II) oxide using hydrogen.

If the atom economy for this reaction is 78% and 4.2 g of copper is formed, calculate the total mass of the reactants. [2]

Mass of reactants = g

10



SECTION B

Answer **all** questions.

7. (a) The boiling temperatures of the hydrides of some Group 4 and Group 7 elements are shown in the table below.

Group 4 hydride	Boiling temperature / °C	Group 7 hydride	Boiling temperature / °C
CH ₄	–161	HF	20
SiH ₄	–112	HCl	–85
GeH ₄	–88	HBr	–66

Hydrogen has an electronegativity value of 2.1.

The electronegativity values of the Group 4 and Group 7 elements are given below.

Element	Electronegativity	Element	Electronegativity
C	2.5	F	4.0
Si	1.8	Cl	3.0
Ge	1.8	Br	2.8



For the hydrides of the Group 4 and Group 7 elements shown in the table opposite

- describe any trends and anomalies in boiling temperature
- explain any differences in boiling temperature in terms of the intermolecular forces present.

[6 QER]

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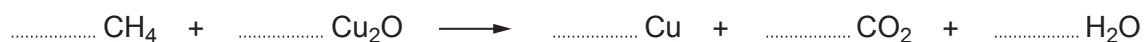
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- (b) Methane reacts with copper(I) oxide according to the following equation.



- (i) Balance the equation. [1]

- (ii) Explain why this reaction is described as a redox process. [1]

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- (c) Compounds containing a Group 1 metal, a Group 3 metal and hydrogen only are known as complex metal hydrides. There has been much interest in their use as hydrogen storage systems for future fuel cell-powered vehicles.

One such hydride contains 38.7% Li and 50.1% Al.

Find the empirical formula of this compound. [2]

Empirical formula



8. (a) Aluminium reacts with oxygen to form aluminium oxide.

Using outer electrons only, draw a dot and cross diagram to show the bonding in aluminium oxide.

[2]

- (b) Aluminium also reacts with chlorine to form aluminium chloride.

A 0.400 g sample of aluminium chloride was heated to 220 °C. The vapour produced occupied a volume of 60.8 cm³ at a pressure of 101 kPa.

Show that the molecular formula of aluminium chloride in the vapour is Al₂Cl₆.

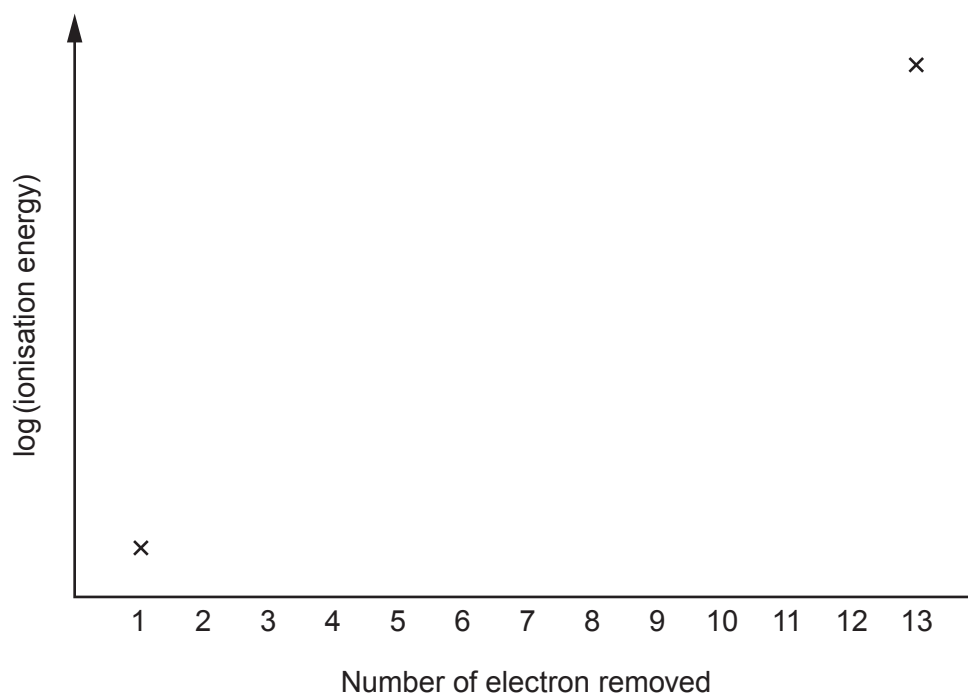
[4]



- (c) (i) Sketch a graph of $\log(\text{ionisation energy})$ for the successive ionisations of aluminium.

The first and last points have been plotted for you.

[1]



- (ii) I. Explain the general slope of the graph.

[1]

.....

.....

.....

- II. Explain the reason for any sharp changes in the graph.

[1]

.....

.....

.....



- (d) Although aluminium has 25 known isotopes only two of them occur naturally. These are ^{27}Al which is stable and ^{26}Al which is radioactive.

^{26}Al decays by electron capture and its half-life is 7.2×10^5 years.

- (i) Give the mass number and symbol of the species produced when an atom of ^{26}Al decays. [1]

.....

- (ii) If 8.0 mg of ^{26}Al decays by electron capture, calculate the mass in **grams** of ^{26}Al left after 2.88×10^6 years. [2]

Mass = g

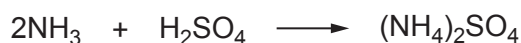
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9. (a) Over 180 million tonnes of ammonia are manufactured each year.

The main use of ammonia is in the production of salts such as ammonium sulfate, which is used as a fertiliser.



Explain why this is an acid-base reaction.

[1]

- (b) Sodium hydroxide reacts with ammonium sulfate to form ammonia, sodium sulfate and water as shown in the equation below.



A 1.86 g sample of ammonium sulfate was neutralised by exactly 26.70 cm³ of a sodium hydroxide solution.

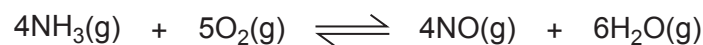
Calculate the concentration, in mol dm⁻³, of the sodium hydroxide solution used.

[3]

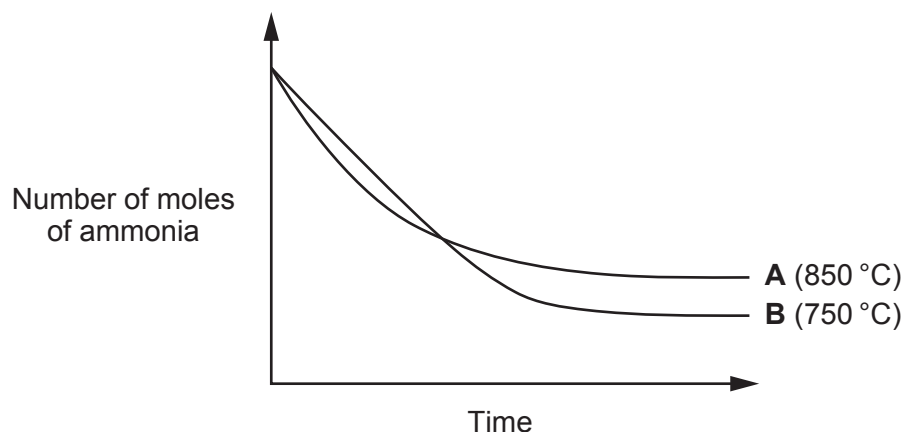
Concentration = mol dm⁻³



- (c) Another use of ammonia is in the production of nitric acid. In the first part of this process ammonia is oxidised in air.



The graph below shows how the number of moles of ammonia present changes as the reaction proceeds until equilibrium is reached.



- (i) State Le Chatelier's principle.

[1]

- (ii) A student looked at curves **A** and **B** and said that the forward reaction is exothermic.

Is he correct? Justify your answer by using Le Chatelier's principle.

[2]

- (iii) On the graph, draw a curve that represents the reaction at 850 °C but with a catalyst added to the reaction mixture.

Label this curve **C**. Explain the shape of the curve.

[2]



- (d) A solution of nitric acid has a concentration of $0.0550 \text{ mol dm}^{-3}$.

Calculate its pH.

[1]

pH =

- (e) A student said that the bonds in an ammonia molecule are not purely covalent.

Explain why she is correct.

[2]

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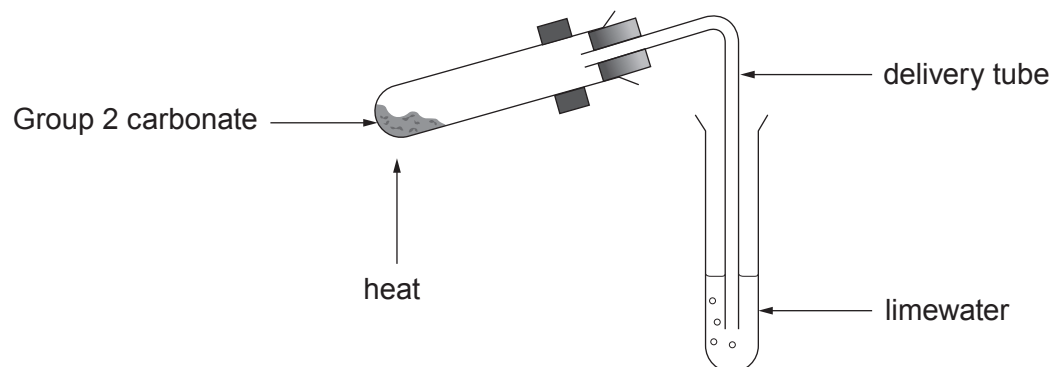
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10. (a) A student investigated the thermal stability of Group 2 carbonates. She used the following apparatus and method.



1. Start a stopwatch at the moment you begin to heat the carbonate and continue to heat for 4 minutes or until the limewater turns cloudy.
2. After the limewater turns cloudy lift the delivery tube out of the limewater then remove the flame from under the boiling tube.
3. Repeat the heating procedure for each carbonate in turn.
4. Use a spatula-measure of the appropriate carbonate, fresh limewater and heat with the hottest Bunsen burner flame each time.

Her results are shown below.

Carbonate	Time taken for limewater to turn cloudy/s
MgCO_3	20
CaCO_3	40
SrCO_3	230
BaCO_3	does not turn cloudy



- (i) Suggest an improvement to the method to ensure that the experiment is a fair test. [1]

.....

.....

- (ii) Suggest why the delivery tube should be lifted out of the limewater before the flame is removed from under the boiling tube. [1]

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.....

- (iii) State what conclusion she can draw about the thermal stabilities of the Group 2 carbonates from these results. [1]

.....

.....

- (iv) The student was told that the temperature at which barium carbonate decomposes is 1360°C . The maximum temperature of a typical Bunsen burner flame is around 800°C .

State whether the limewater would have turned cloudy if she had used two Bunsen burners to heat the barium carbonate. Give a reason for your answer. [1]

.....

.....

.....



- (b) Barium nitrate also decomposes on heating.



- (i) In an experiment 0.960 g of barium nitrate was heated strongly for 2 minutes.

Calculate the maximum volume, in cm^3 , of gas that could be produced at a temperature of 25°C and a pressure of 1 atm.

[3]

Volume = cm^3

- (ii) The volume of a gas is directly proportional to its temperature at constant pressure.

A student said that if the gas formed in this experiment were collected at a temperature of 50°C and at 1 atm pressure, the volume formed would be double that calculated in part (i).

Do you agree? Justify your answer.

[1]

.....
.....



- (c) State the conditions necessary for **each** of barium oxide and barium metal to conduct electricity. Explain this property in terms of structure and bonding in each case. [3]

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- (d) The atomic radius of a barium atom is 0.217 nm.

From the list below, choose the value for the **ionic** radius of a barium ion. Give a reason for your choice. [2]

0.135 nm

0.210 nm

0.217 nm

0.265 nm

.....

.....

.....

- (e) A sample of barium contains two isotopes. The first isotope has a relative isotopic mass of 134.9 and the second a relative isotopic mass of 137.9. The relative atomic mass of the sample of barium is 137.3.

Calculate the percentage abundance of the first isotope. [2]

Abundance = %

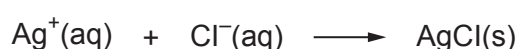


11. Seawater contains a number of dissolved salts. Although composition varies with location, $1\,000\text{ cm}^3$ of seawater contains about 20 g of chloride ions, Cl^- , and about 3 g of sulfate ions, SO_4^{2-} .

A student is given a sample of seawater from Rhossili Bay and asked to determine the chloride ion content by volumetric analysis and the sulfate ion content by gravimetric analysis.

- (a) Determination of chloride ion content by volumetric analysis.

The method is similar to an acid-base titration. A silver nitrate solution of known concentration is used to precipitate chloride ions as silver chloride.



The seawater is diluted by a factor of five before it is used in the titration.

The endpoint of this titration is difficult to determine directly, so potassium chromate(VI), K_2CrO_4 , is used as an indicator. When all of the chloride ions have been used up, the chromate(VI) ions react with silver ions and produce silver chromate(VI), which forms a red precipitate. The instant a permanent red tinge appears in the solution, the endpoint has been reached.

Volume of diluted seawater in the conical flask = 25.0 cm^3

Concentration of silver nitrate solution in the burette = 0.100 mol dm^{-3}

Mean titre = 26.40 cm^3

- (i) Before starting the titration, the student rinses the burette with silver nitrate solution. Suggest why he does this. [1]

.....

.....

- (ii) Suggest why the student dilutes the seawater. [1]

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- (iii) Describe how the student should dilute the seawater by a factor of five. [3]

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- (iv) Describe and explain **one** action the student might take just before the endpoint of the titration, to ensure that the volume of silver nitrate added at the endpoint is accurate. [2]

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- (v) Write an ionic equation for the precipitation of silver chromate(VI). [1]

.....

- (vi) Calculate the mass of chloride ions in $1\,000\text{ cm}^3$ of the original seawater, giving your answer to an **appropriate** number of significant figures. [4]

Mass of chloride ions = g



- (b) Determination of sulfate ion content by gravimetric analysis.

100 cm³ of undiluted seawater and 0.100 mol dm⁻³ barium nitrate solution were used.

The mass of the barium sulfate precipitate was 0.65 g.

You may assume that **all** of the sulfate ions in the seawater were precipitated.

- (i) Describe how the student carried out the gravimetric analysis to find the mass of the barium sulfate precipitated. [5]

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- (ii) Calculate the minimum volume, in cm^3 , of barium nitrate solution needed to precipitate all of the sulfate ions in 100cm^3 of the seawater.

[3]

Volume = cm^3

- (iii) Suggest why the volume of barium nitrate needed was different to the volume of seawater used.

[1]

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END OF PAPER

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